Ideological polarization in climate change discourse: an NLP analysis of YouTube political commentators

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Abstract

The project investigates how political commentators on YouTube discuss and frame the issue of climate change. By analyzing transcripts from both progressive and conservative channels, it examines how language reflects ideological divides, differing beliefs, and tones of debate. The goal is to uncover patterns of polarization, identify shared and contrasting narratives, and explore how political orientation shapes the way climate change is portrayed in online discourse.

1 Introduction

1.1 Past works

Recent studies show that conversations about climate change online are becoming increasingly tied to political identity. On platforms like X (Twitter), people often use language that reflects the views of their political group [3]. Other work [1] demonstrates that political leaning can even be detected automatically from how people or news talk about issues. This political division is especially visible in climate related discussions; some research [5], for example, find strong emotional and ideological polarization in online climate conversations, while others [2] report that this divide has been growing over time on social media. Overall, studies suggest that climate discussions online are shaped not only by environmental concerns but also by deeper political identities, making polarization a key aspect to understand how information spreads on social platforms.

1.2 Extracting the transcripts

YouTube has become one of the most influential spaces for political communication, where ideological narratives are often amplified by algorithms and personality-driven content. Compared to traditional media, political discourse on YouTube tends to be more direct, polarized, and linguistically diverse, making it an ideal setting to study ideological framing around hot topics like climate change.

The data used in this project comes entirely from YouTube videos, which have been selected using the official *YouTube API* and whose transcripts have been extracted using the unofficial *YouTube Transcript API*. All the selected videos come from either famous American YouTube political commentators or from American politicians, and videos were selected to ensure balanced representation between progressive and conservative perspectives, where the speaker "general ideology" is defined by either a party affiliation or established reputation of the YouTuber.
In particular, the criteria that I used to select the videos to analyze are the following:

- The video should have the least number of speakers, in order to minimize the noise; ideally, the video should have only one speaker.
- The speaker should have a clear political stance, or be a politician belonging either to the Republican party or to the Democratic party.
- In the video, there must be a discussion on climate change.

Based on these criteria, I was able to extract 34 videos from 14 different speakers, that, based on their political affiliation or general content of their YouTube channel, could be classified like this:

Speaker	General ideology	Total minutes	Videos
Secular Talk	Progressive	66:52	9
David Pakman	Progressive	45:50	7
Democrat politicians	Progressive	43:39	1
Ben Shapiro	Conservative	26:41	3
Candace Owens	Conservative	10:32	2
Donald Trump	Conservative	9:22	2
Joe Rogan	Conservative	7:54	1
Jordan Peterson	Conservative	7:18	1
Charlie Kirk	Conservative	4:15	1
Vivek Ramaswamy	Conservative	4:02	2
Republican politicians	Conservative	3:03	2
John Coleman	Conservative	2:39	1
Dan Pena	Conservative	2:24	1
Fox News	Conservative	2:18	1

Table 1: Overview of the speakers

1.3 Text cleaning pipeline

After being extracted, each transcript has gone through a pipeline, in order to prepare the data for the following analysis. The following steps have been done:

1. Each text has been first split into chunks of maximum 250 words, to balance semantic coherence and computational manageability.

2. Since automatic YouTube captions come without punctuation, I used a pretrained model for automatic punctuation restoration (Deep Multilingual Punctuation).

3. YouTube videos are often noisy and heterogeneous, containing digressions or off-topic segments. To mitigate this issue, I implemented a semantic filtering step: I used *Cosine Similarity* to compare the embeddings of each text chunk (obtained with a pretrained model from Hugging Face) to a set of 10 chosen words related to climate change. Then, I filtered to keep only the chunks in which at least one of the 9 concepts related to climate change has a similarity greater than 0.4 with the given text chunk.

2 Research question and methodology

The general goal of this project is to try to understand the characteristics of the language used by YouTube political commentators when talking about climate change, and try to detect how ideological orientation influences their linguistic framing when talking about this topic.
Based on this goal, a key part of the task was to label the data. An easy solution would have been to use the manual labels determined by the general ideology of the channel (1), however I wanted to know if language models are able to detect speakers' stance without explicit ideological labels. The solution that I chose was to use a method known as *Zero-Shot Classification*, which classifies data into categories it has never seen before during training, therefore without requiring any labeled examples of those categories. This can be obtained by using the general language understanding of the model (in my case I used a pretrained model called Political_DEBATE_base_v1.0, specific for tasks regarding political texts). This step effectively provided a weak supervision signal; the key idea behind this is that labels must be decided beforehand, and the model will assign to each chunk the label with the highest confidence.
The labels that I chose, inspired by the most recurrent dichotomy in climate discourse, alarmist vs. skeptical narrative, are the following:

- "Believes climate change is an urgent crisis";
- "Believes climate change is exaggerated".

3 Experimental results

3.1 Stance detection

This step represents a shift toward a more context-sensitive approach, building upon the previously generated labels to develop a tailored classifier.
Text embeddings were created using a general-purpose model (all-MiniLM-L6-v2), to test the efficiency of general-purpose models on specific tasks related

Speaker	**Chunks aligned with general ideology**
Democrat Politicians	17/17 (100%)
Republican politicians	3/3 (100%)
Jordan Peterson	3/3 (100%)
Fox News	2/2 (100%)
Vivek Ramaswamy	2/2 (100%)
John Coleman	2/2 (100%)
Charlie Kirk	1/1 (100%)
Dan Pena	1/1 (100%)
Secular Talk	32/34 (94%)
Candace Owens	5/6 (84%)
Joe Rogan	3/4 (75%)
David Pakman	21/30 (70%)
Ben Shapiro	11/20 (55%)
Donald Trump	1/2 (50%)

Table 2: Coherence of zero-shot labels with the general speaker ideology ("progressive" or "conservative").

to politics. The produced embeddings were then fed into a supervised learning model trained to recognize stance patterns specific to the linguistic and rhetorical style of the speakers under study.
Dimensionality reduction was performed using *Principal Component Analysis* on the obtained text embeddings (reduced to 50 dimensions), followed by *Logistic Regression* on trained on 80% of the data, with hyperparameters selected via 5-fold *grid search* according to accuracy.

	precision	recall	f1-score	support
False	0.76	0.94	0.84	17
True	0.80	0.44	0.57	9
accuracy			0.77	26
macro avg	0.78	0.69	0.71	26
weighted avg	0.78	0.77	0.75	26

Figure 1: Classification report of the Logistic Regression applied on the test set after splitting the data, in order to find the best hyperparameters.

Next, the classifier was retrained using the best hyperparameters on the entire dataset to assign final stance labels to all text chunks. Although the classification report on the full dataset was not used for model evaluation, since it measures fit quality rather than generalization performance, it showed a notable increase in F1-score for both classes, indicating that the model successfully captured the overall linguistic patterns of the corpus.

Task	Used model
Create embeddings for zero-shot classification	Political_DEBATE_base_v1.0
Perform zero-shot classification to obtain chunks labels	Political_DEBATE_base_v1.0
Create embeddings for stance detection	all-MiniLM-L6-v2
Perform stance detection	Logistic Regression

Table 3: Recap of used models until now.

This approach allows the model to generalize beyond the initial zero-shot outputs, refining and contextualizing the stance predictions according to the actual distribution of opinions within the corpus. It can be considered a more reliable and interpretable representation of how climate change is framed across different speakers and contexts, moving from broad semantic inference to evidence-based stance modeling.
Logistic Regression was selected as the stance classifier due to its interpretability and robustness on moderately sized datasets. Unlike more complex neural models, logistic regression provides clear decision boundaries and allows direct examination of how different embedding features contribute to stance prediction.

3.2 Overall speakers similarity

An interesting thing that can be done is analyzing how similar each speaker is compared to each other. To do so, I used the embeddings created for the stance detection, and for each speaker I computed a representative embedding vector, obtained by using the mean of all the embedding vectors of each speaker. This allows us to calculate the similarity not at the text chunk level, but at an aggregate level, comparing the whole speaker stance with each other.
In particular, the similarity is, once again, based on the *cosine similarity* between the given vectors. Because of the nature of the conversations, most of the transcripts are overall quite similar; however, the implementation of an effective cleaning pipeline and consequent embedding model, allows the similarity to be very truthful in terms of results, as can be seen in the Figure 2. In fact, as we can see, speakers that have a general progressive ideology tend to be close to each other, and the same happens with the conservatives. Also, we can notice that democrat politicians and republican politicians have a high similarity; my interpretation is that the language used by politicians is in general quite similar, despite the ideas being different, while YouTube commentators (like Jordan Peterson or Charlie Kirk, located on the opposite side of the graph) might have a more informal and direct language, which might not be typical of official political debates.

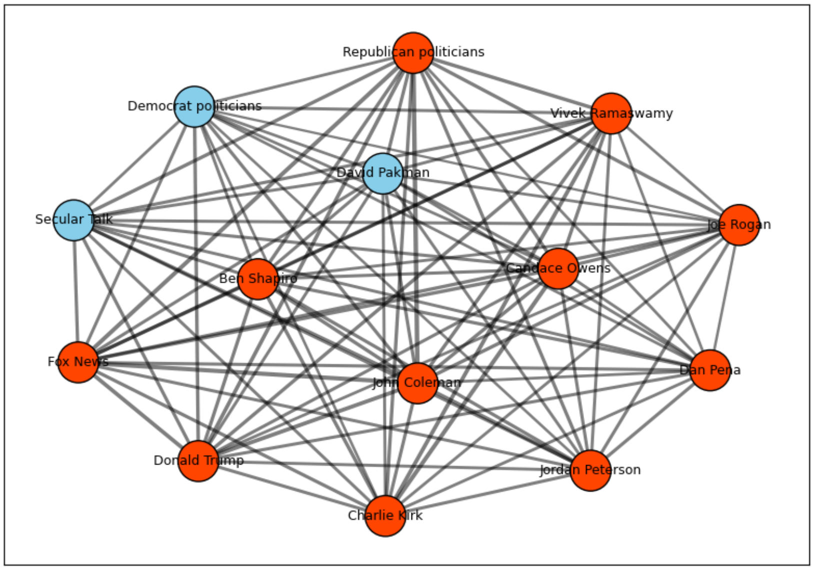

Figure 2: Cosine similarity between speakers.

3.3 Clustering

Another way of analyzing the similarity between text is by using a clustering method, which groups together similar elements. Once again, I used the embeddings that I created before, and I decided to perform dimensionality reduction on these embeddings by using the *UMAP* method, reducing the number of dimensions to 2 so that the results can be easily visualized on a 2D plot.
Next, I used *K-Means clustering* to perform the actual clustering on the reduced data; I used $k = 2$, since I wanted to split the data into two clusters, each ideally representing one of the two stances. The plot can be seen in Figure 3.

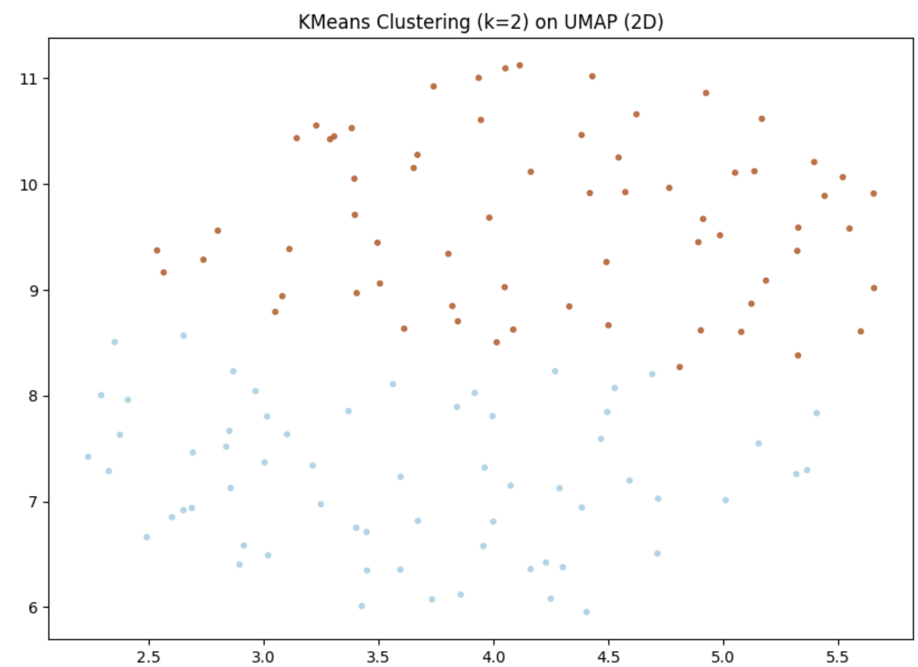

Figure 3: K-Means clustering

If we group the data points by the stance determined previously with the logistic regression model, we have a proof that the clustering was performed in a correct way. As we see in Figure 4, in fact, the text chunks belonging to cluster "1" are likely to be chunks of speakers considered more "progressive", while the text chunks belonging to cluster "0" are definitely those belonging to the more "conservative" speakers.

lr_believes_exaggeration	kmeans_cluster	count
False	1	57
	0	38
True	0	25
	1	8

Figure 4: Text chunks grouped by the previously detected stance, with the associated k-means cluster.

3.4 Named-entity recognition

One thing I that I consider useful when working on text data is to analyze the entities mentioned by the speakers. In my case, I used a pretrained model *(en_core_web_sm)* to extract the *named entitities* from the speakers' transcripts, and looked at the differences between the mentioned entities, grouped by predicted class; here are the results:

- Chunks of speakers classified as *"Believes climate change is an urgent crisis"*:
  - Interesting mentioned people: B. Obama, A. Camus (philosopher), Leonardo DiCaprio (actor) and B. Sanders.
  - Mentioned various times China, Florida, Middle East, Paris (likely because of the Paris climate agreement), and many states/geographical areas.
- Chunks of speakers classified as *"Believes climate change is exaggerated"*:
  - Multiple mentions of Al Gore (former politician), Bill Gates and the Bible.
  - Very few mentions of geographical entities; some news channels (CNN, Fox) mentioned.

3.5 LIME explanations

After running a ML model, it is fundamental to be able to explain the meaning behind the results, and understand how to interpret them. I decided to use *LIME*, an algorithm that can explain the predictions of any classifier or regressor in a faithful way, by approximating it locally with an interpretable model [4]. In my specific case, the aim is to understand which words in each text chunk push my logistic regression model toward one of the two classes. What LIME does is, for each chunk, it generates a number of perturbed samples (400 in my case) by randomly removing or altering words, and the model's predictions on those samples are analyzed to estimate how each word contributes to the classification.
For example, if we take the words that are considered to be most influentials, grouped by predicted stance, we have the results shown in Tables 4 and 5.

Word	**Average word magnitude**
Energy	0.462
Fuels	0.393
Oil	0.242

Table 4: The words that influence the model to classify a chunk into the "Believes climate change is an urgent crisis" stance.

In the case of chunks classified as "Believes climate change is an urgent crisis" we can see that words that the words influencing the most the predictions are those related to pollution.

Word	**Average word magnitude**
Hoax	0.592
Science	0.391
People	0.373

Table 5: The words that influence the model to classify a chunk into the "Believes climate change is exaggerated" stance.

In this case, instead, the most influential word is "hoax", which is not strange at all. It is interesting to notice also that if, instead of using the mean of each word magnitude we consider the median, in the case of people that believe the climate change is exaggerated the top 3 influential words are "religion", "Gates" and "skeptic".

4 Concluding remarks

This study explored how ideological polarization shapes climate change discourse among YouTube political commentators. By combining zero-shot classification, stance detection, and clustering, it was possible to move from weak supervision signals to a more context-aware understanding of each speaker's framing. YouTube videos are inherently noisy, with varied production quality, spontaneous speech, and informal language, which complicates text analysis.
The analysis revealed clear differences between progressive and conservative commentators in both linguistic patterns and referenced entities. Today, the main division is not between believers and skeptics (outright denial is rare) but rather between those who view climate change as an urgent crisis and those who downplay its immediacy.
Logistic Regression on text embeddings provided an interpretable model of stance, robust on a moderately sized dataset, while clustering and named-entity recognition highlighted recurring themes and references characteristic of each stance.
Overall, this project shows the effectiveness of combining NLP techniques with interpretable models to study ideological framing online. Beyond climate change, the methodology offers a scalable framework for analyzing polarization and stance across topics and platforms.

Parts of this project hve been developed with the assistance of OpenAI's ChatGPT (GPT-5). The AI was used to support the development of project ideas, the structuring of methodological workflows, the creation of the paper abstract, and the identification of relevant references. All content produced with AI assistance has been carefully reviewed, edited, and validated by me. I take full responsibility for the final content and its accuracy, relevance, and academic integrity. In addition, Google Gemini has been used to generate some pictures used in the presentation slides.

References

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